

Types of Languages

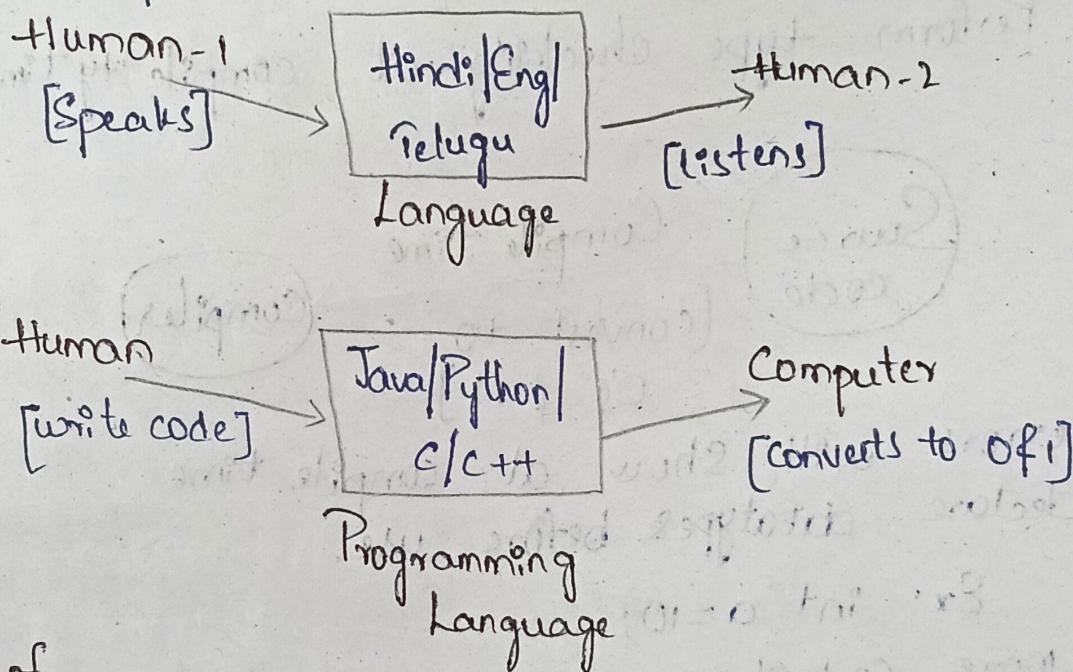
1. Procedural:

Introduction to Programming Language

* Computers understand only binary language i.e. 0's and 1's at very minute level.

What is programming language?

- Is a formal language used to write instructions that a computer can understand and execute.



Types of programming Languages:

1. Procedural:

→ Series of well-structured steps & procedures to compose a program.

→ Contains a systematic order of statements, functions and commands to complete a task.

2. Functional:

→ Writing a program only in pure functions i.e., never modify variables but only create new ones as an output.

→ Used in a situation where we have to perform lots of different operations on the same set of data like ML.

3. Object-Oriented:

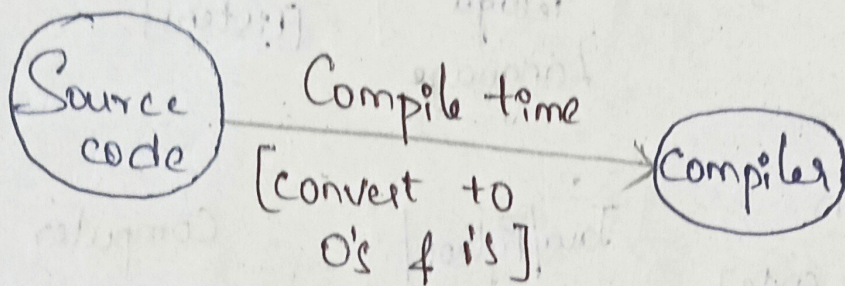
→ Revolves around the objects

→ Code + data = 'objects' (Data + Methods)

→ developed to make it easier to develop, debug, reuse and maintain.

Static Languages:

→ Perform type checking at compile time.



→ errors will show at compile time

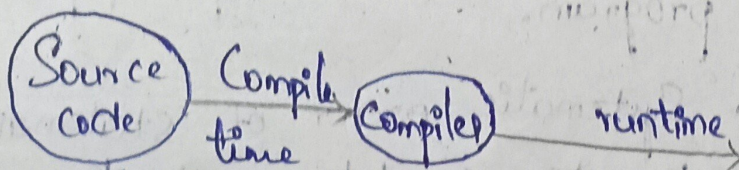
→ declare datatypes before use.

Ex: `int a=10;`

→ more control over the program.

Dynamic Languages:

→ Perform type checking at run-time.



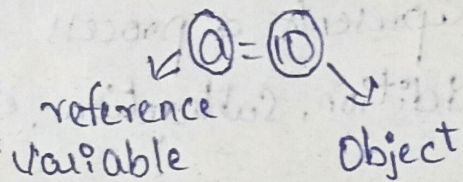
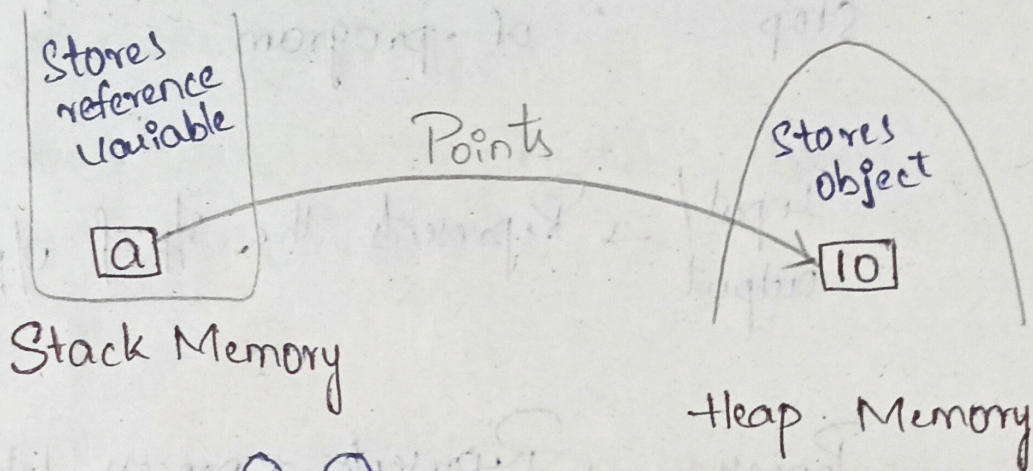
→ error might not show till the programs run.

→ no need to declare datatype of variables

Ex: `a=10;` [language by itself figures out datatype]

→ Saves time in writing code but might give error at runtime.

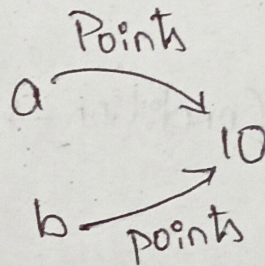
Memory Management



Ex:

`a = 10;`

`b = a;`



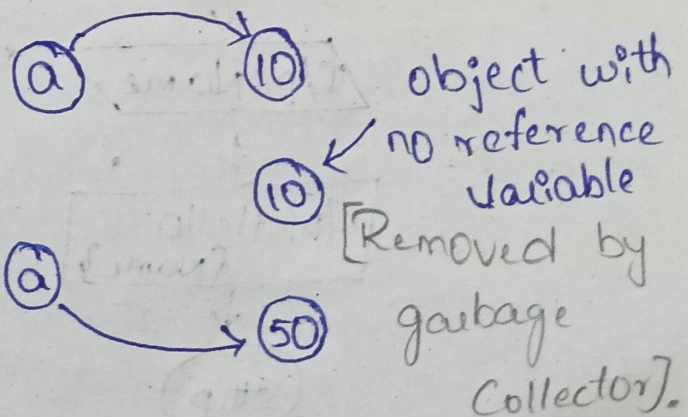
→ more than one reference variable can point towards one object.

→ If any of the reference variable changes the Object then it is changed for all reference variables that points towards same object.

Ex:

Initially,

`a = 10`

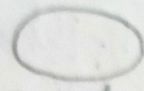


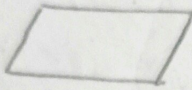
After,

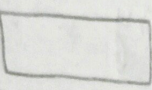
`a = 50`

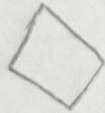
Flow of Program

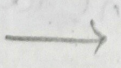
Flow chart Symbols:

1.  oval → Start/Stop → Represents start or stop of program.

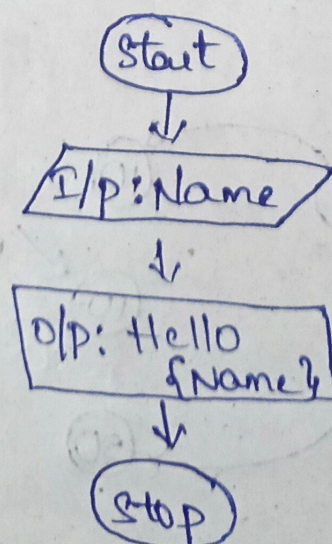
2.  parallelogram → Input/output → Represents the i/p & o/p.

3.  Rectangle → Processing → Represents a process like addition, subtraction, etc...

4.  Diamond/Rhombus → Condition → Represents for conditional statements

5.  Arrow → Flow direction of program → A line connector which shows what is the flow of program.

Example: Take a name and output Hello Name:



Pseudo Code:

- It is just a way to write steps which is human readable format. [It is not a code].
- It is mainly meant for human reading not for machine reading.

Ex: Take above example to take a name & output Hello name:

Step-1 : Start

Step-2 : Take input from user [name=input
("enter name")]

Step-3: Hello {Name} [output]

Step-4: Stop.